Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
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EXPERIMENT NUMBER: 6

AIM: To analyze network traffic using the Wireshark tool. Perform following activities:

- Packet capture and filtering
- Analyze ethernet frames
- IP packet analysis
- ICMP analysis (PING test)
- HTTP/HTTPs request analysis
- Analyze header of TCP /UDP dump

THEORY:

1. What is Wireshark? Explain Packet Capture and Filtering.

Wireshark is a widely used network protocol analyzer that allows users to capture, inspect, and analyze data packets travelling through a computer network. It provides detailed information about network communication, including the protocols used, source and destination addresses, packet contents, and timing information. Wireshark is useful for network troubleshooting, performance analysis, protocol analysis, and educational purposes.

Packet capture is the process of collecting network packets that are transmitted or received through a network interface such as Wi-Fi or Ethernet. In Wireshark, the user selects an active network interface and starts the capture. The captured packets are displayed in real time, with information such as packet number, time, source address, destination address, protocol, length, and additional details.

Packet filtering is used to view only the packets that are relevant to a particular analysis. Wireshark provides display filters that allow users to isolate packets belonging to specific protocols or conditions.

Examples of display filters are:

- `ip` – Displays IP packets.
- `icmp` – Displays ICMP packets.
- `tcp` – Displays TCP packets.
- `udp` – Displays UDP packets.
- `http` – Displays HTTP packets.
- `tls` – Displays TLS packets used in secure communication.

Thus, packet capture collects network traffic, while filtering helps in analyzing specific packets from the captured data.

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2. What is an Ethernet Frame? Explain its Main Fields.

An **Ethernet frame** is a unit of data transmitted over an Ethernet network. When data is sent through a local area network, it is encapsulated inside an Ethernet frame before transmission. The Ethernet frame contains addressing and control information required for delivering data from one device to another within a network.

The main fields of an Ethernet frame are:

1. **Preamble:**
   It consists of a sequence of bits used to synchronize communication between the sender and receiver.
2. **Start Frame Delimiter (SFD):**
   It indicates the beginning of the actual Ethernet frame.
3. **Destination MAC Address:**
   This field contains the physical address of the receiving device. It is generally 48 bits long.
4. **Source MAC Address:**
   This field contains the physical address of the sending device and is also generally 48 bits long.
5. **Type/Length:**
   This field identifies the type of protocol contained in the data field, such as IPv4 or IPv6. In some Ethernet frame formats, it indicates the length of the data.
6. **Data or Payload:**
   This field contains the actual data being transmitted. It may contain an IP packet or data from another network-layer protocol.
7. **Frame Check Sequence (FCS):**
   This field is used for error detection. The receiving device checks the FCS to determine whether the frame was damaged during transmission.

Therefore, an Ethernet frame provides the necessary physical addressing and error-checking information for communication over an Ethernet network.

3. What is an IP Packet? Explain the Important Fields of its Header.

An **IP packet** is a unit of data that is transmitted across a network using the Internet Protocol (IP). The IP packet contains a header and a payload. The header contains information required to identify, route, and deliver the packet from the source device to the destination device.

The important fields of an IPv4 header are:

1. **Version:**
   Specifies the version of the Internet Protocol being used. For IPv4, the version number is 4.
2. **Internet Header Length (IHL):**
   Specifies the size of the IP header.
3. **Type of Service / DSCP:**
   Provides information related to packet priority and quality of service.

4. **Total Length:**
   Specifies the total size of the IP packet, including both the header and data.
5. **Identification:**
   Used to identify fragments belonging to the same original packet.
6. **Flags:**
   Used to control the fragmentation of packets.
7. **Fragment Offset:**
   Indicates the position of a fragment within the original packet.
8. **Time To Live (TTL):**
   Specifies the maximum number of network hops that a packet can pass through. The TTL value is reduced by routers, preventing packets from circulating indefinitely.
9. **Protocol:**
   Identifies the upper-layer protocol contained in the packet. Examples include TCP, UDP, and ICMP.
10. **Header Checksum:**
    Used to detect errors in the IP header.
11. **Source IP Address:**
    Specifies the IP address of the sending device.
12. **Destination IP Address:**
    Specifies the IP address of the receiving device.

Thus, the IP header contains essential information that allows packets to be identified, routed, and delivered across networks.

4. What is ICMP? Explain PING Using ICMP.

ICMP (Internet Control Message Protocol) is a network-layer protocol used for sending error messages, status information, and diagnostic messages between network devices. Unlike TCP and UDP, ICMP is not generally used for transferring application data. Instead, it helps in diagnosing network connectivity and reporting communication problems.

One of the most common applications of ICMP is the **PING**

command. When a user enters the following command:

ping google.com

the following process takes place:

1. The source computer sends an **ICMP Echo Request** packet to the destination.
2. The packet travels through the network toward the destination device.
3. If the destination is reachable and configured to respond, it sends back an **ICMP Echo Reply** packet.
4. The source computer receives the reply and calculates the response time.
5. This process may be repeated several times to check connectivity and packet

loss. Using Wireshark, the packets can be observed by applying the following display

filter:

icmp

The user can then identify the **Echo Request** and **Echo Reply** packets.

Therefore, PING uses ICMP messages to test whether a remote host is reachable and to measure the time required for communication between the source and destination.

5. Differentiate Between HTTP and HTTPS

HTTP	**HTTPS**
HTTP stands for **Hypertext Transfer Protocol**.	HTTPS stands for **Hypertext Transfer Protocol Secure**.
It is used for communication between a web browser and a web server.	It provides secure communication between a web browser and a web server.
Data is transmitted without encryption.	Data is encrypted using TLS/SSL.
It is less secure for transmitting sensitive information.	It is more secure for transmitting sensitive information.
It generally uses **Port 80**.	It generally uses **Port 443**.
It does not provide strong protection against interception or modification of data.	It provides encryption and helps protect the confidentiality and integrity of data.
The URL generally begins with `http://`.	The URL generally begins with `https://`.

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6. Explain the Main Fields of TCP and UDP Headers.

TCP Header Fields

TCP (Transmission Control Protocol) is a connection-oriented and reliable transport-layer protocol. Its header contains several important fields:

1. **Source Port:**
   Identifies the application or process sending the data.
2. **Destination Port:**
   Identifies the application or process receiving the data.
3. **Sequence Number:**
   Specifies the sequence position of data and helps maintain the correct order of transmission.
4. **Acknowledgment Number:**
   Indicates the next sequence number expected from the sender.
5. **Header Length:**
   Specifies the size of the TCP header.
6. **Flags:**
   Control the TCP connection. Common flags include SYN, ACK, FIN, and RST.
7. **Window Size:**
   Specifies the amount of data that can be received before an acknowledgment is required.
8. **Checksum:**
   Used for error detection in the TCP segment.
9. **Urgent Pointer:**
   Indicates urgent data when the URG flag is set.

UDP Header Fields

UDP (User Datagram Protocol) is a simple connectionless transport-layer protocol. Its header contains four main fields:

1. **Source Port:**
   Identifies the sending application.
2. **Destination Port:**
   Identifies the receiving application.
3. **Length:**
   Specifies the total length of the UDP header and data.
4. **Checksum:**
   Used for error detection.

The UDP header is smaller and simpler than the TCP header, resulting in lower communication overhead.

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7. Differentiate Between TCP and UDP

TCP	UDP
TCP stands for **Transmission Control Protocol**.	UDP stands for **User Datagram Protocol**.
It is a connection-oriented protocol.	It is a connectionless protocol.
It establishes a connection before transmitting data.	It sends data without establishing a connection.
It provides reliable data delivery.	It does not guarantee reliable data delivery.
Lost packets can be retransmitted.	Lost packets are generally not retransmitted by UDP itself.
It uses acknowledgments.	It does not use acknowledgments.
It provides sequencing to maintain the correct order of data.	Packets may arrive out of order.
It provides flow control and congestion control.	It has lower overhead and does not provide these mechanisms in the same way.
The TCP header is larger and more complex.	The UDP header is smaller and simpler.
It is generally slower because of reliability mechanisms.	It is generally faster because of lower overhead.

Examples include web communication, email, and file transfer.	Examples include DNS, online gaming, streaming, and real-time communication.

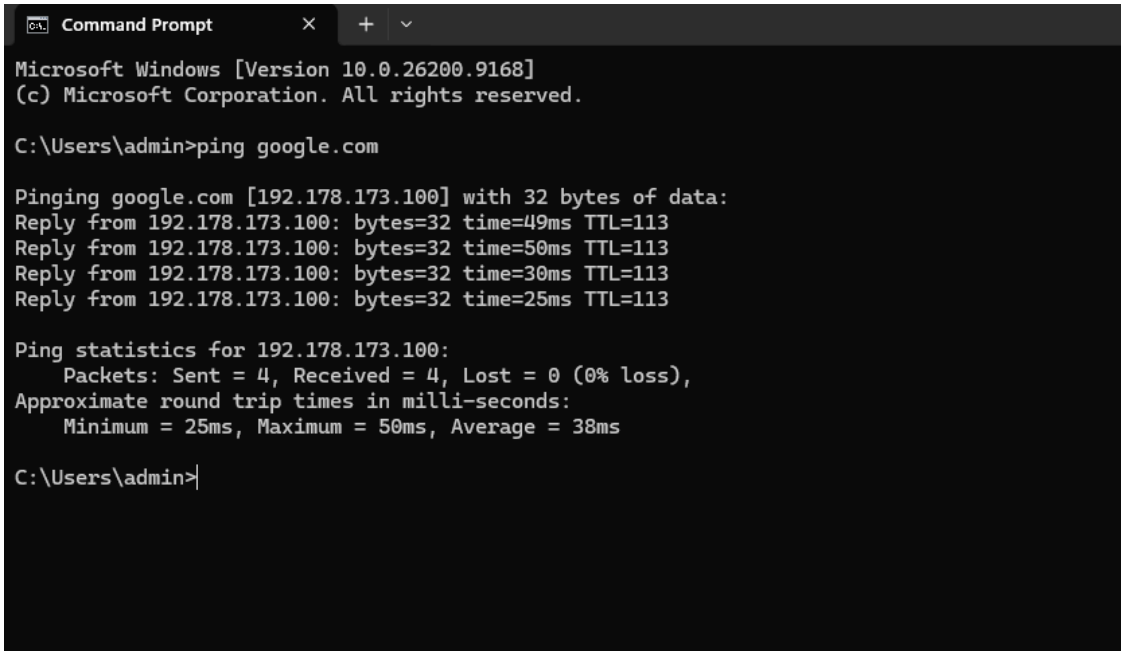
PROCEDURES:

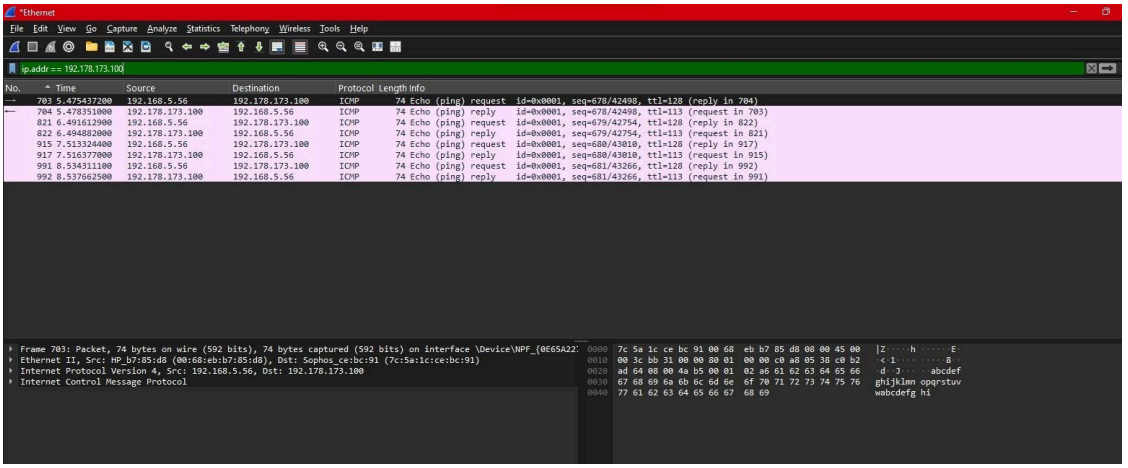
1. Open Wireshark and select the active network interface such as Wi-Fi or Ethernet.
2. Start packet capture and generate network traffic by opening a website or using the command ping google.com.
3. Stop the capture after sufficient packets are collected.
4. Apply display filters such as ip, icmp, tcp, udp, http, or tls to view specific packets.
5. Select a packet and expand Ethernet II to observe source MAC address, destination MAC address, and frame type.
6. Expand Internet Protocol Version 4 (IPv4) to analyze source IP, destination IP, TTL, protocol, and packet length.
7. Apply the icmp filter and analyze PING Echo Request and Echo Reply packets.
8. Apply the http or tls filter to examine HTTP/HTTPS requests and TLS handshake information.
9. Apply the tcp filter and analyze TCP header fields such as source/destination ports, sequence number, acknowledgment number, and flags.
10. Apply the udp filter and analyze UDP header fields such as source/destination ports, length, and checksum.
11. Record the observations and conclude the network traffic analysis.

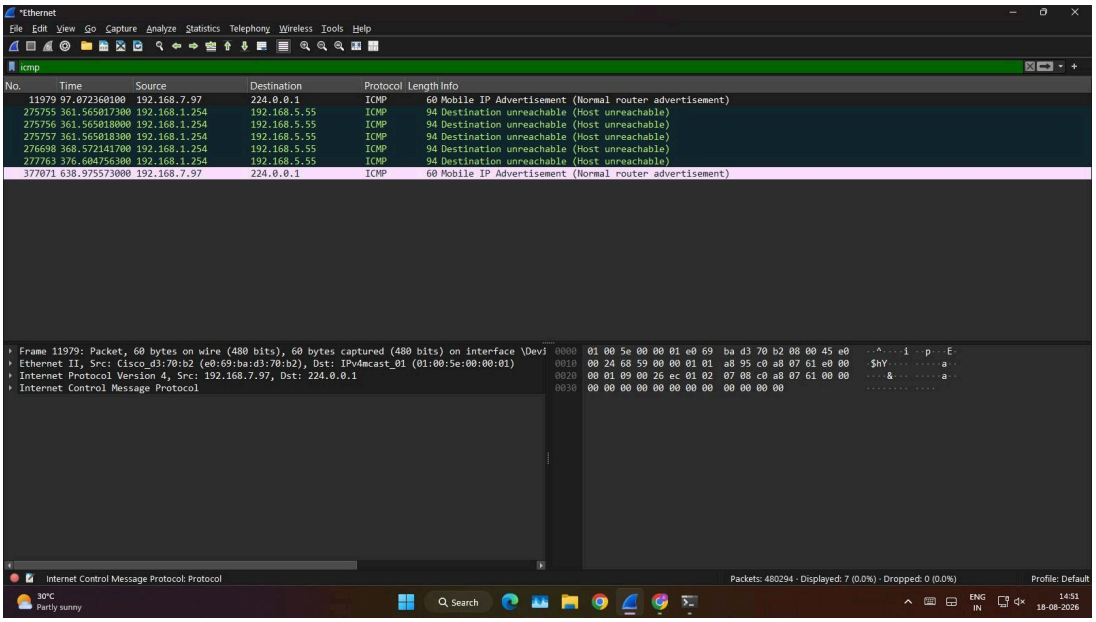
RESULT:

The network traffic was successfully captured and analyzed using Wireshark. Different protocols including Ethernet, IPv4, ICMP, TCP, UDP, HTTP, and TLS were identified using display filters. The header fields and communication details of different packets were successfully observed and analyzed.

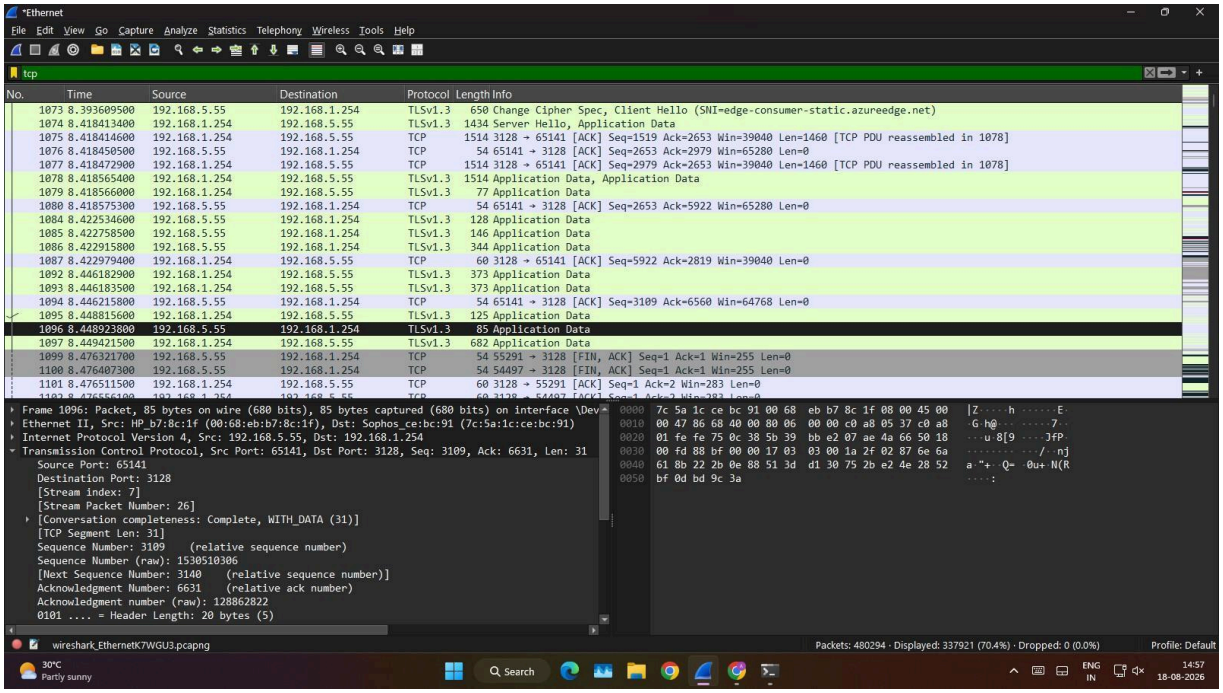
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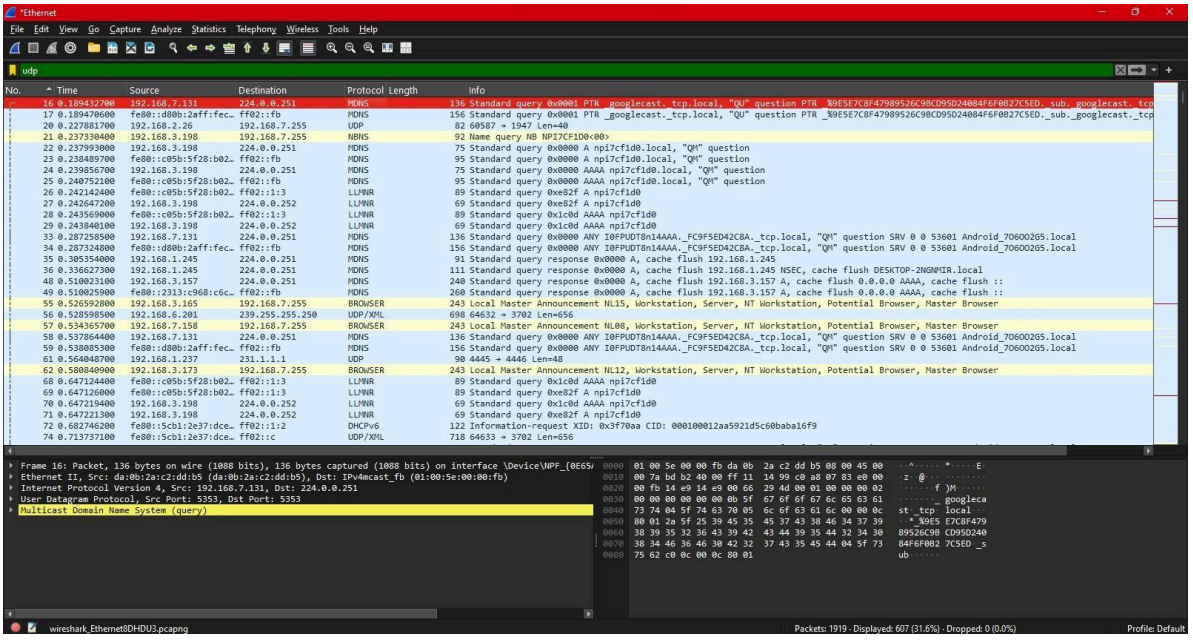




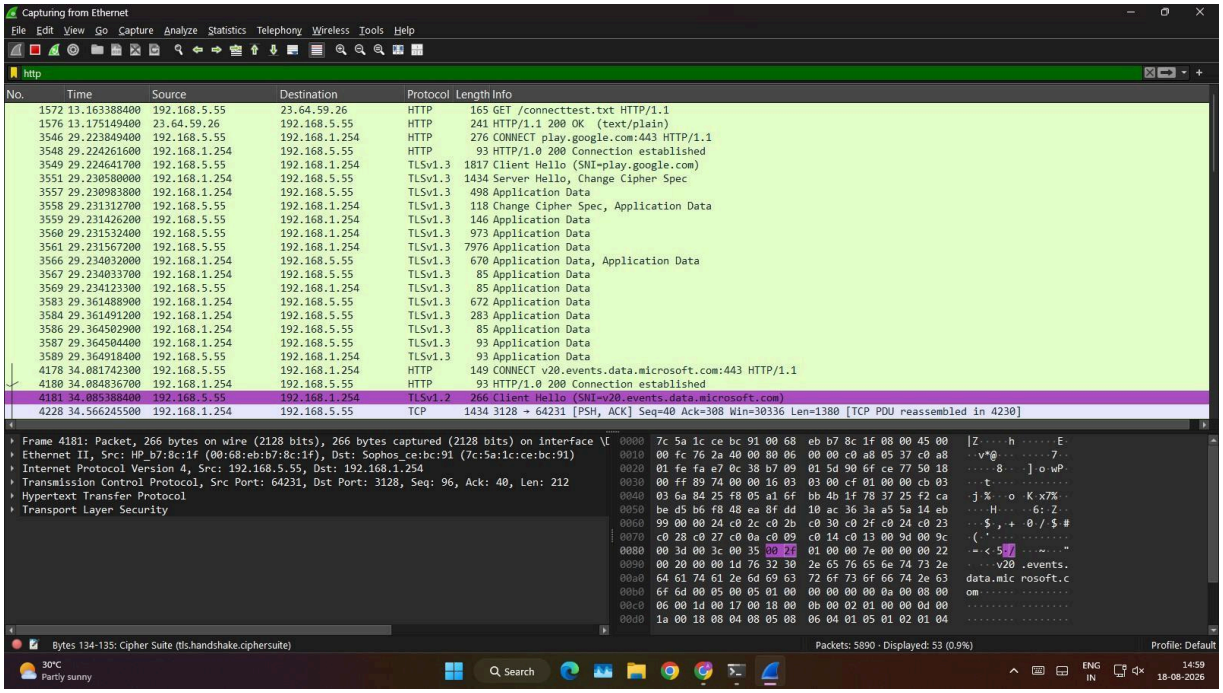


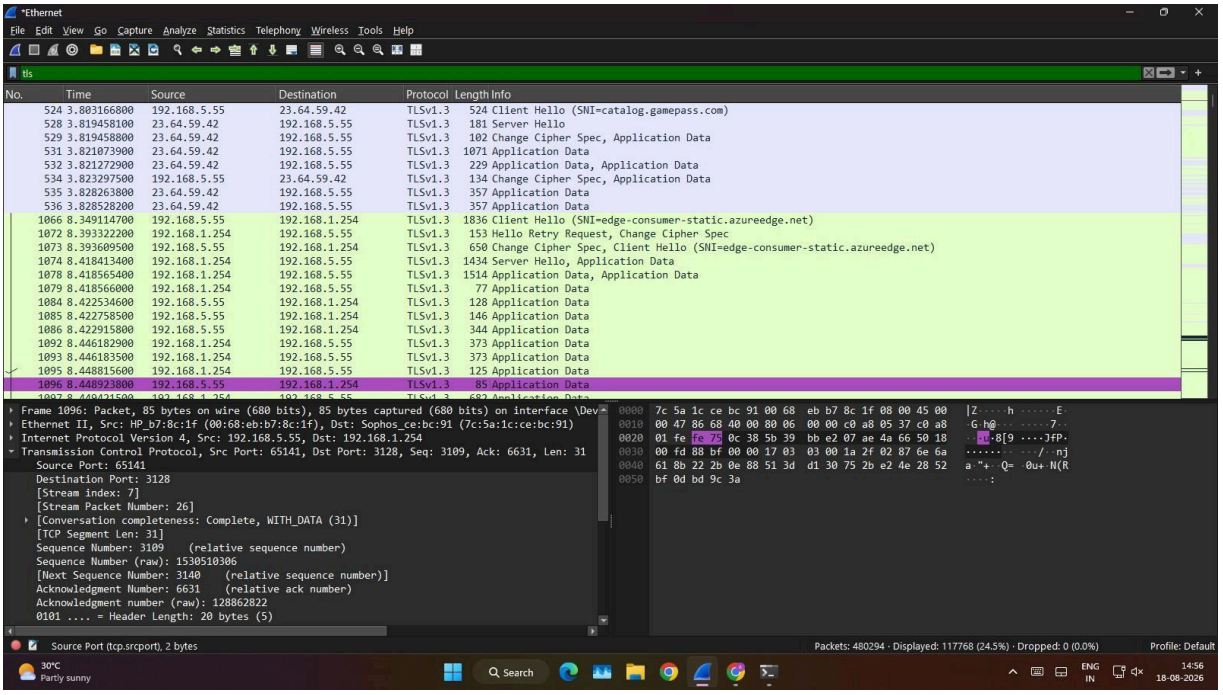
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CONCULSION:

Thus, the network traffic was successfully analyzed using the Wireshark tool. Packet capture and filtering were performed, and Ethernet frames, IP packets, ICMP PING packets, TCP and UDP headers, and HTTP/HTTPS communication were examined. The experiment helped in understanding how different network protocols communicate and how packet-level information can be captured and analyzed using Wireshark.